Identifying the historical quartic input set

Human Compression project: a component of the L059 recovery

The numerical work of De Saedeleer, Leemans, Mixer and Pisanski [1] uses the census of edge-transitive tetravalent graphs of Wilson and Potočnik. The census has since been expanded and its identifiers have changed. Here we identify the set of graph isomorphism classes required for a corrected version of that calculation. We use an explicit coverage assertion of the census authors, together with finite checks of the retained data. We do not reconstruct the meaning of every historical identifier.

All graphs are finite, simple and connected. A tetravalent graph has degree four at each vertex. For a connected bipartite graph X, let $F = \mathrm{Aut}(X)$ and let $A = \mathrm{Aut}^+(X)$ be the subgroup preserving each part of its bipartition. A graph is *dart-transitive* if F is transitive on ordered adjacent pairs; it is *semisymmetric* if F is edge-transitive but not vertex-transitive. Define the property P by

$$|V(X)| \le 150, \qquad X \text{ bipartite}, \qquad A \text{ edge-transitive}, \qquad |F_v| < 200. \tag{1}$$

In this setting the order of F_v is independent of the choice of v. In the semisymmetric case the two vertex orbits are the two parts, which have equal size because the graph is regular.

Lemma 1. *Among bipartite edge-transitive graphs, A is edge-transitive exactly in the dart-transitive and semisymmetric cases. In these cases*

$$\frac{|A|}{|V(X)|} = \frac{|F_v|}{2}.$$

Proof. If F is not vertex-transitive, its two vertex orbits are the two parts. Thus $A = F$ is edge-transitive and orbit–stabilizer gives $|F| = (|V(X)|/2)|F_v|$. If F is dart-transitive, any edge may be sent to another with its black endpoint sent to the black endpoint. Such an automorphism belongs to A. Here $[F : A] = 2$, and $|F| = |V(X)|\,|F_v|$.

In the remaining case F is vertex- and edge-transitive but not dart-transitive. If A were edge-transitive, its point stabilizer would be transitive on the four neighbours of a point. Every automorphism fixing a point preserves the parts, so $A_v = F_v$. Vertex-transitivity of F would then imply dart-transitivity, a contradiction. The two stated formulas follow from the two orbit–stabilizer calculations. □

The acquired premises. We retain two historical copies of the census table, archived on 5 May 2008 and 28 December 2014. The historical homepage describes the table as the list of graphs in the census. As with any use of this census, we acquire the authors' assertion that its entries represent distinct isomorphism classes and that the historical graph invariants are correct. We do not claim an independent reconstruction of every historical graph.

The later primary source [2, §1, physical p. 2] describes the expanded census as

> "a collection of all known edge-transitive graphs of valence 4 up to 512 vertices."

We acquire precisely the resulting coverage premise: every graph in either retained historical census occurs, up to isomorphism, in the later census at the linked site. These historical graphs were already known before the 2016 statement. The retained current homepage cites

that paper, links the full Magma data, and explains that the older identifiers changed when the search was reorganized. The files, URLs and hashes are recorded in `inputs.json` and `source_acquisition.json`.

This is a coverage premise about graphs already in an earlier catalogue. The authors explicitly do not assert completeness among all semisymmetric tetravalent graphs. We acquire no such stronger classification.

Finite verification. The checker parses every row of each of the three tables and every graph occurrence in the full Magma source. The results are:

Edition	Table rows	Rows satisfying P	Dart-transitive	Semisymmetric
5 May 2008	788	397	272	125
28 December 2014	793	397	272	125
Current full census	7364	397	272	125

There are 807 graphs of order at most 150 in the current data, of which 562 are bipartite. All 562 are checked to be connected, simple, tetravalent and pairwise nonisomorphic, using canonical nauty certificates. Their full automorphism groups, edge orbits, vertex orbits and, when needed, dart orbits are computed. Each computed automorphism is also checked directly against the edge set. The resulting symmetry type, full group order and vertex stabilizer order agree with the current table in every case. Exactly 397 satisfy (1). Thus the current graph count is checked on actual edge sets, independently of the printed count.

Theorem 1. *Under the stated historical census and coverage premises, either retained historical census and the retained current census contain the same 397 graph isomorphism classes satisfying P.*

Proof. Let $\mathcal{H}$ be the set of classes satisfying P in one of the historical editions, and let $\mathcal{C}$ be the corresponding set in the current census. The acquired historical assertion and the parsed row count give $|\mathcal{H}| = 397$. The computations on current edge sets give $|\mathcal{C}| = 397$.

Every graph representing a class in $\mathcal{H}$ was known before the expanded census and has at most 150 vertices, hence at most 512. The coverage premise places an isomorphic copy in the later census. Property P is preserved by graph isomorphism: an isomorphism conjugates the full automorphism groups, carries the two bipartition classes to the two classes, and preserves the orders and orbit conditions in (1). Thus $\mathcal{H} \subseteq \mathcal{C}$. Since the sets have the same finite cardinality, $\mathcal{H} = \mathcal{C}$. The argument applies separately to the two historical editions. □

Scope and attribution. The threshold in (1) is explicit. The original paper says that the vertex stabilizer has order less than 100. That is not the present threshold: by the lemma, (1) is instead equivalent to $|\operatorname{Aut}^+(X)|/|V(X)| < 100$. The original paper also explicitly includes $K_{4,4}$, whose full vertex stabilizer has order 144. Thus its stated cutoff and its stated example already disagree. The present theorem identifies the historical input set for the corrected cutoff; it neither silently changes the original statement nor establishes why the discrepancy arose.

The graph identification theorem concerns uncoloured isomorphism classes. It does not fix an ordering of the two parts, transfer a particular historical identifier, or certify the later marking enumeration. Those are separate tasks. The census construction, historical invariants and coverage assertion are borrowed from Wilson and Potočnik. Our contribution here is the explicit finite comparison and the short set argument permitting reuse of the current edge data. No new classification or solution of an open problem is claimed.

References

[1] J. De Saedeleer, D. Leemans, M. Mixer and T. Pisanski, *Core-free, rank two coset geometries from edge-transitive bipartite graphs*, Mathematica Slovaca **64** (2014), 991–1006; arXiv:1106.5704v2.

[2] S. Wilson and P. Potočnik, *Recipes for Edge-Transitive Tetravalent Graphs*, arXiv:1608.04158v1 (2016), §1; accompanying full census.

[3] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation **60** (2014), 94–112. The finite checker uses nauty through the pynauty interface; its installed version is recorded in the receipt.